

Harsh Goel

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Research focus: My research develops the foundations for proactive video assistants that seek relevant information, reason about events and human intent, and coordinate with people and other agents.

Education

Ph.D. in Computer Engineering, The University of Texas at Austin, Austin, TX (2023–2027 (expected))

M.S. in Robotics, University of Pennsylvania, Philadelphia, PA (2021–2023)

B.Eng. in Mechanical Engineering and Computer Science, National University of Singapore, Singapore (2016–2020)

Research Experience

The University of Texas at Austin | Graduate Research Assistant, Swarm Lab | Austin, TX | 2023–present

- **Video agents:** Developed VSeek using temporal-logic rubric rewards for long-video search post-training, achieving relative gains of up to 8% in Pass@1 and 15% in Pass@4 over base models (ECCV 2026 oral).
- **Activity understanding:** Developed self-distillation with Bayesian feedback for theory-of-mind reasoning from video, supporting intent inference and activity anticipation (*From Actions to Intent*, under review).
- **Multi-agent RL:** Developed R3DM, linking learned roles to predicted future behavior; achieved higher test win rates than prior role-based methods on selected SMAC/SMACv2 tasks (ICML 2025).

Google DeepMind | Research Scientist Intern | Mountain View, CA | 2025

- **LLM search agents:** Developed a stabilized GRPO variant and synthetic data pipeline for multi-hop question answering; raised out-of-domain performance by 9.7 percentage points versus the same training setup without synthetic data.

NVIDIA | Machine Learning Research Intern | Santa Clara, CA | 2024

- Trained planning models to initialize classical search-based planners with learned priors, reducing optimization time by 23% for low-speed autonomous driving maneuvers.

University of Pennsylvania | Research Assistant, Kumar Lab | Philadelphia, PA | 2021–2023

- Developed and open-sourced hierarchical RL algorithms for aerial vehicles (M.S. thesis; advisor: Prof. Vijay Kumar). IROS 2022 | Code for Skill Learning | Video.

National University of Singapore | Deep Learning Researcher | Singapore | 2018–2021

- **Multi-agent learning:** Developed SocialLight and CoordLight for decentralized traffic control; contributed graph neural network and attention models for distributed multi-agent learning using Ray, PyTorch, and TensorFlow.
- **Distributed multi-agent learning for search:** Explored evolutionary algorithms and reinforcement learning for robotic information gathering (advisor: Prof. Guillaume Sartoretti). Evolutionary Planning | MARL Search.
- **Robotic assembly under imperfect vision:** Investigated grasping, teleoperation, and assembly (advisor: Prof. Marcelo H. Ang Jr.). Perception and Control Code | Teleoperation | Video.

Selected Peer-Reviewed Publications

* Equal contribution.

- Incentivizing Vision Language Models to Search for Long Video Question Answering. **Goel, Harsh**; Sharan, SP; Shah, Sahil; et al. In *European Conference on Computer Vision (ECCV; oral presentation)*, 2026.
- CrossView: Can Vision-Language Models Reason Across Cameras? Shah, Sahil*; Sharan, SP*; **Goel, Harsh***; et al. In *European Conference on Computer Vision (ECCV)*, 2026.
- NeuS-QA: Grounding Long-Form Video Understanding in Temporal Logic and Neuro-Symbolic Reasoning. Shah, Sahil*; Sharan, SP*; **Goel, Harsh***; et al. In *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- Neuro-Symbolic Evaluation of Text-to-Video Models Using Formal Verification. Sharan, SP; Choi, Minkyu; Shah, Sahil; **Goel, Harsh**; Omama, Mohammad; Chinchali, Sandeep. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- R3DM: Enabling Role Discovery and Diversity Through Dynamics Models in Multi-Agent Reinforcement Learning. **Goel, Harsh**; Omama, Mohammad; Chalaki, Behdad; Tadiparthi, Vaishnav; Moradi Pari, Ehsan; Chinchali, Sandeep. In *International Conference on Machine Learning (ICML)*, 2025.
- Distributed Upload and Active Labeling for Resource-Constrained Fleet Learning. Akcin, Oguzhan; **Goel, Harsh**; Zhao, Ruihan; Chinchali, Sandeep P. In *Conference on Robot Learning (CoRL)*, 2025.
- CoordLight: Learning Decentralized Coordination for Network-Wide Traffic Signal Control. Zhang, Yifeng; **Goel, Harsh**; Li, Peizhuo; Damani, Mehul; Chinchali, Sandeep; Sartoretti, Guillaume. *IEEE Transactions on Intelligent Transportation Systems*, 2025.

- Towards Neuro-Symbolic Video Understanding. Choi, Minkyu; **Goel, Harsh**; Omama, Mohammad; Yang, Yunhao; Shah, Sahil; Chinchali, Sandeep. In *European Conference on Computer Vision (ECCV; oral presentation)*, 2024.
- SocialLight: Distributed Cooperation Learning towards Network-Wide Traffic Signal Control. **Goel, Harsh**; Zhang, Yifeng; Damani, Mehul; Sartoretti, Guillaume. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023.

Selected Preprints and Manuscripts

- Self Distillation Q Learning. **Goel, Harsh**; Xu, Haoran; Zhang, Amy; Chinchali, Sandeep P. Under Review, 2026.
- Revisiting On-Policy Self-Distillation from an RL Perspective. Xu, Haoran; Ma, Huidi; **Goel, Harsh**; Zhang, Amy. Under Review, 2026.
- T-LESSA: Temporal-Logic Event Specification Steering of Video Diffusion Attention. Choi, Minkyu; **Goel, Harsh**; Homsy, Soamar; Chinchali, Sandeep P. Under Review, 2026.
- From Actions to Intent: Self-Distillation with Bayesian Feedback Enables Theory-of-Mind Reasoning from Video. **Goel, Harsh**; Joarder, Shourov; Dubey, Aditya; Chinchali, Sandeep P. Under Review, 2026.
- CPRD: Contrastive Predictive Role Discovery for Multi-Agent Reinforcement Learning. **Goel, Harsh**; Lee, Dongsu; Tadiparthi, Vaishnav; Moradi Pari, Ehsan; Mahjoub, Hossein Nourkhiz; Zhang, Amy; Chinchali, Sandeep P. Under Review, 2026.
- Bayesian Partner Modelling Enables Adaptive Replanning for LLM Coordination. **Goel, Harsh**; Ellendula, Aditya Sai; Tadiparthi, Vaishnav; Moradi Pari, Ehsan; Mahjoub, Hossein Nourkhiz; Chinchali, Sandeep P. *arXiv preprint arXiv:2608.18490*, 2026.
- S3-RL: Learning to Retrieve and Answer Step-by-Step with Synthetic Data. **Goel, Harsh**; Udathu, Akhil; Jabireddy, Susmija; Kalkar, Pradnesh; Parulekar, Atharva. *arXiv preprint arXiv:2605.01248*, 2026.

Work Experience

MoveIAI | Robotics Software Engineer | Singapore | 2020

- Engineered planning algorithms, SLAM, and visualization tools for construction-site robots.

Thales Alenia Space | Attitude and Orbit Control Engineer | Singapore | 2019

- Developed low-earth orbit attitude control algorithms for satellites under sensor-constrained state estimation.

Agency for Science, Technology and Research | Robotics Engineering Intern | Singapore | 2018

- Developed real-time tracking and interception software for edge robots. Code.

Past Projects

Synthetic Data Curation for Image-Language and Time Series Foundation Models (Jan 2025–May 2025)

Steered diffusion models to generate synthetic data that reduces uncertainty in downstream model predictions. Code.

Improving Image Captioning for Visual Localization for Robotics (May 2024–Dec 2024)

Optimized in-context examples with DSPy to improve image captions for text-image robot localization. Code.

Language-Guided Multi-Agent Reinforcement Learning (Aug 2023–Mar 2024)

Used LLMs to guide exploration and improve sample efficiency in multi-agent RL. Code | Video.

Safe Imitation Learning for Autonomous Vehicles (F1-Tenth) (Jan 2023–May 2023)

Devised **safe visual imitation learning** algorithms based on barrier functions in **ROS 2** for autonomous driving. Code.

Honors and Awards

- Cockrell School of Engineering Graduate Research Fellowship, UT Austin (2023)
- GAPSA Travel Grant, University of Pennsylvania (2023)
- Dean's List (top 5%), 2017–2020
- IEEE Hackathon, 2nd place (2019)
- NUS Hackers Night, 1.5K award (2018)

Service

Conference reviewer: ICLR (2027); AAAI, COLM, CVPR, CoRL, ECCV, ICML, and NeurIPS (2026); CoRL (2025).

Technical Skills

Programming: Python, C/C++, Java, MATLAB, JavaScript. **Tools:** Docker, Kubernetes, Git.

Machine learning and robotics: PyTorch, JAX, TensorFlow, Scikit-learn, Ray, ROS 2, OpenCV.